

MODUL UNIVERSITY VIENNA

SEID 2364

SESSION 9

Spatial Distance, Authority, and Risk

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*Mediation Pathways in
Distributed AI Systems*



Purpose: Establish continuity. This is not a new topic, but a new lens.

Speaking Notes:

"We are continuing the same system from last week, but looking at it from a different angle. Last week we built pathways. Today we stress them."

Instructions

- Work individually in your journal
- Do not discuss yet
- Use your Assignment 8 system as reference
- Focus on your own reasoning

2

Purpose: Establish a private baseline.

Speaking Notes:

"Do not optimize your answers. I want your current assumptions. Use your existing system—do not invent a new one."

List the potential benefits and harms this system can produce.

- Be sure to consider that there are multiple types of harms/benefits
- These may differ across stakeholders (individual users, creators, organizations, and the platform).

Instructor Notes:

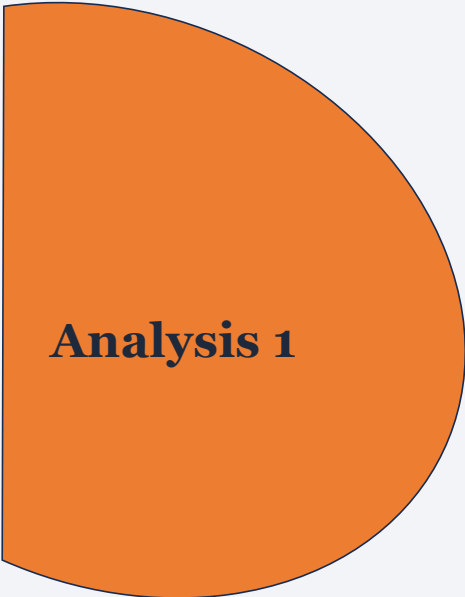
Purpose:

- Surface intuitive judgments
- Establish baseline variation in reasoning

Expect:

- Broad, unsystematic answers
- Mixing of harms, benefits, and assumptions

Do not correct yet—this variability is necessary for the next steps.



Analysis 1

In the system you have already analyzed using Assignment 8,

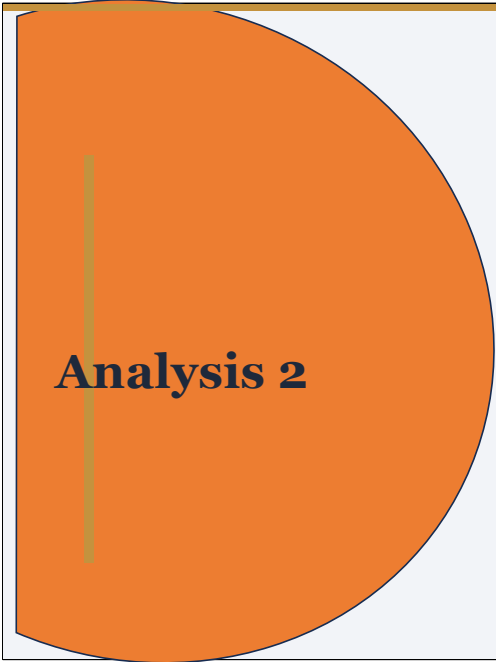
- identify where the signal exists in physical space and virtual space at different stages of the pathway.
- Where does it transition between these spaces, and what changes when it does?

3

Purpose: Activate prior BBS thinking.

Speaking Notes:

"Work silently. Do not define 'signal.' Just describe what you think is happening."



Analysis 2

- In your previous analysis, where did the system operate locally, and where did it operate remotely?
- How did those spatial differences affect control, visibility, or responsibility, and how might that change the ethical evaluation of the system?
- **From whose perspective is the system local or remote?**

4

Purpose: Layer spatial reasoning and introduce observer-dependence without formalizing it.

Speaking Notes:

"Add to your answer—do not erase. Where does your system stop being local?"

"Your answer may change depending on where you are in the system."



Analysis 3

- Does increasing spatial distance in a mediation pathway increase ethical risk?
- Why or why not?
- **If possible, provide at least one example where distance increases risk and one where it does not.**

5

Purpose: Force generalization.

Speaking Notes:

"Justify your answer. Provide both a confirming case and a counterexample."

What You Just Did

You traced systems

You made assumptions about space

You evaluated risk → Without being told how to do it

7

Purpose: Reveal implicit assumptions. This slide is a metacognitive pause—it names what students just did without realizing it. The goal is to make visible the reasoning strategies they used intuitively so these can be examined, questioned, and refined in subsequent activities.

Key Pedagogical Points:

- "You traced systems" — Students mapped mediation pathways from Assignment 8 without formal instruction on system tracing. Highlight that they already possess this analytical capability; the course is giving it structure and vocabulary.
- "You made assumptions about space" — In Analyses 1 and 2, students distinguished local from remote operations. Most will not have questioned whether "local" and "remote" are stable categories. This is the opening to introduce observer-dependence more formally later.
- "You evaluated risk without being told how" — This is the critical reveal. Students performed ethical risk assessment

using intuitive heuristics. The arrow notation on the slide emphasizes that this happened implicitly. The next section (Flight Path Test) will show why intuitive risk evaluation can be unreliable when spatial conditions change.

Speaking Notes:

"You evaluated risk assuming the system works. Did anyone consider what happens when it doesn't?"

"Notice that nobody asked you to evaluate risk—you did it automatically. That tells us something important: you already have a risk model in your head. The question is whether it's the right one."

"Most of you assumed spatial proximity equals control. We're about to test that assumption."

Transition to Next Slide:

Use this moment to pivot from reflection to challenge. The Flight Path Test (next slide) will destabilize the assumptions surfaced here by introducing variable access conditions. Let the class sit with the discomfort of having reasoned without a framework before moving on.

Timing:

Spend approximately 2–3 minutes on this slide. Do not rush—the pause is pedagogically important. Allow a brief moment of silence after the opening question before continuing with the additional speaking points.

The Flight Path Test

What changes when access to a system varies across space?

7A

Purpose: Introduce real-world variability.

Speaking Notes:

"As I move from ground to flight, access changes—full, partial, none. The system has not changed, but what it can do has changed."

"We are not modeling meaning. We are modeling whether the signal remains usable."

From Assignment 8 → Today

You already identified:

Bias entry

Transformation

Amplification

Now add: → Where does distance change?

8

Purpose: Bridge prior work.

Speaking Notes:

"Same system, new lens. Where does the signal degrade?"

Group Task

Identify in your system:

- Entry point of bias
- Transformation point
- Amplification point
- Spatial transition

9

Purpose: Structured synthesis.

Speaking Notes:

"Work in small groups. Identify concrete locations in your system."

System Breakdown

1. Entry

2. Transformation

3. Amplification

4. **Spatial Shift**

10

Purpose: Shared schema.

Speaking Notes:

"Now add: where does the signal degrade?"

Ethics as War

What strategic thinking reveals about evaluating systems

Purpose: Section transition. Signal a shift in register—we are leaving the opening provocations and entering a more structured analytical frame. The title “Ethics as War” is deliberately provocative. It should land as a question, not a statement.

Speaking Notes:

(Let the slide sit for a moment before speaking.)

“We have been looking at how systems use space—how distance, separation, and mediation change the relationship between a system and the people it affects. Now we are going to sharpen the tools we use to evaluate those systems.”

“And we are going to borrow those tools from an unexpected place: military strategy.”

(Pause. Let them read the subtitle.)

“Why war? Because strategists have spent centuries thinking about a very specific problem: how do you make decisions when you cannot trust the information you have? When the other side is actively trying to mislead you? When conditions on the ground are changing faster than your plans can adapt?”

“That problem—deciding under uncertainty, with incomplete and possibly deceptive signals—turns out to be exactly the problem we face when we try to evaluate complex systems. The systems we analyzed in Assignment 8 do not come with honest labels. They do not announce their full capabilities. They operate in conditions where oversight degrades and information is incomplete.”

“So for the next few minutes, we are going to sit with two thinkers who changed how the world understands conflict. And then we are going to ask: what do their ideas mean for how we evaluate the systems that make decisions about people?”

Transition: Advance to Sun Tzu slide.



c. 544 – 496 BC
Military strategist, ancient China

Sun Tzu

Chinese general, strategist, and philosopher whose influence on Eastern and Western military thinking has endured for over two millennia.

The Art of War, composed around the 5th century BC,

Oldest known work on military strategy and remains required reading in war colleges worldwide.

Core insight: the highest form of victory is won **before battle is ever joined**—through deception, intelligence, and the manipulation of what your adversary believes to be true.

For Sun Tzu, **deception is not a failure of ethics**—it is the foundation of strategy. Intent is something to be **concealed, manufactured, and weaponized**.

Picture source: <https://www.britannica.com/biography/Sunzi>

9A

Purpose: Introduce the first strategic thinker in the “Ethics as War” section. Establish that deception—the deliberate concealment and manufacture of intent—is foundational to strategic thinking. This sets up the Clausewitz slide, which will then ask: if intent is unreliable, what do you evaluate instead?

Speaking Notes:

“We are going to spend a few minutes with two thinkers who changed how the world understands conflict. And then we are going to ask what their ideas mean for the systems you have been analyzing.”

“Sun Tzu wrote *The Art of War* around 500 BC. It is the oldest surviving work on military strategy, and it is still assigned at West Point, Sandhurst, and every serious war college in the world. Twenty-five centuries and it has not been replaced. That should tell you something.”

“His most famous line: All warfare is based on deception. Think about what that means. He is not saying deception is a tactic you sometimes use. He is saying it is the foundation. The entire structure of strategic advantage rests on controlling what the other side believes.”

(Pause.)

“When you are strong, appear weak. When you are near, appear far. When you are ready to attack, appear unprepared. This is not a footnote in his theory—this is the theory.”

“So here is the question this raises: if the highest form of strategy is built on making your adversary misjudge your intentions, what does that tell us about using intent as a way to evaluate any system that operates strategically?”

(Let them sit with this. Do not answer it yet.)

“Hold that question. We are going to meet someone who took it further.”



c. 544 – 496 BC
Military strategist, ancient China

Sun Tzu

兵者詭道也。

“All warfare is based on deception.”

— Sun Tzu, *The Art of War* (c. 5th century BC)

9B

Purpose: The quote reveal. Let the line land on its own before unpacking it. This is the moment where deception moves from a historical observation to a structural principle.

Speaking Notes:

(Advance the slide. Let them read it. Wait.)

“All warfare is based on deception. Six words. Twenty-five centuries old. Still true.”

“Notice what he is NOT saying. He is not saying deception is useful in warfare. He is not saying it is one tool among many. He is saying it is the foundation. The entire structure of strategic advantage rests on controlling what the other side believes to be true.”

(Pause.)

“Think about what follows from this. If deception is foundational, then intent is not just difficult to read—it is actively constructed to mislead. When you are strong, appear weak. When you are near, appear far. When you are ready to attack, appear unprepared.”

“So Sun Tzu gives us a world where intent is something to be manufactured and weaponized. Not something you can trust. Not something you should base your evaluation on.”

“Now here is where this connects to your work as computer scientists. You are not on a battlefield. But you are in an environment where systems face adversarial threats—where malicious actors are actively trying to deceive your systems, manipulate your inputs, exploit the gap between what a system sees and what is actually there. Adversarial ML, prompt injection, social engineering—these are all forms of what Sun Tzu described.”

“But there is a second front that is even more interesting. Even without a malicious actor, complex systems can become opaque to the people who built them. The system’s internal state becomes hidden—not because anyone is lying, but because the system has grown beyond anyone’s ability to fully observe it. That is also a world where intent becomes unreadable.”

“Sun Tzu described a world where intent is unreliable. The next thinker we meet asked the obvious follow-up: if you cannot trust intent, what do you evaluate instead?”

Transition: Advance to Clausewitz intro slide.

Ethical Framing



“A system that cannot be read cannot be held accountable. Opacity is not a limitation—it is a capability.”

10B

Purpose: The Sun Tzu systems pivot. Translate “all warfare is based on deception” into the language of system accountability. This slide asks: if deception is foundational to strategy, what does that mean for systems whose operations cannot be observed?

Speaking Notes:
(Let them read the slide. Pause.)

“If Sun Tzu were evaluating the systems you analyzed in Assignment 8, this is what he would ask. Not what does the system intend—but can you see what it does? And if you cannot, who benefits from that?”

“Let me give you five ways this plays out in the real world.”

CATEGORY 1: COMPETITIVE DECEPTION

“First, the legitimate version. Every corporation practices strategic opacity. You do not reveal your product roadmap.

You do not show your pricing before launch. Trade secrets exist because information asymmetry is competitive advantage. This is Sun Tzu applied to quarterly earnings—when you are strong, appear weak.”

“Most people accept this as normal. But notice: the system is already designed to be unreadable to outsiders. The question is where the boundary falls between competitive secrecy and public accountability.”

CATEGORY 2: REGULATORY OPACITY

“Second, designing systems that cannot be audited. You do not technically lie to the regulator—you build something complex enough that the regulator cannot ask the right questions. Think about the financial instruments that caused 2008. They were not hidden. They were so structurally opaque that the people selling them could claim they did not fully understand them either. Complexity becomes a shield.”

CATEGORY 3: LIABILITY THROUGH IGNORANCE

“Third—and this is the one I want you to think hardest about—deliberate non-measurement. If you never measure disparate impact, you can never be found to have known about it. If you do not collect the data that would reveal harm, the harm becomes invisible. Not because it does not exist, but because you designed the information architecture to not capture it.”

“This is deception by omission of observation. Sun Tzu would recognize it immediately.”

CATEGORY 4: DARK PATTERNS AND MANUFACTURED CONSENT

“Fourth, user-facing deception. The system presents itself as serving you while actually optimizing for something else. We value your privacy—accompanied by 47 pages of terms nobody reads and an opt-out buried six menus deep. The stated intent is user experience. The actual optimization target is

engagement, data extraction, ad revenue.”

“The system appears to be playing one game while actually playing another. Remember our earlier discussion—this is the equivalent of throwing the game for other purposes.”

CATEGORY 5: THE BLACK BOX DEFENSE

“Fifth, algorithmic opacity as legal strategy. Our algorithm is proprietary. The model is too complex to explain. These are not just technical limitations—they are strategic positions. If you cannot be required to explain how your system makes decisions, you cannot be held accountable for those decisions in the same way. The inscrutability is performing a function.”

SYNTHESIS:

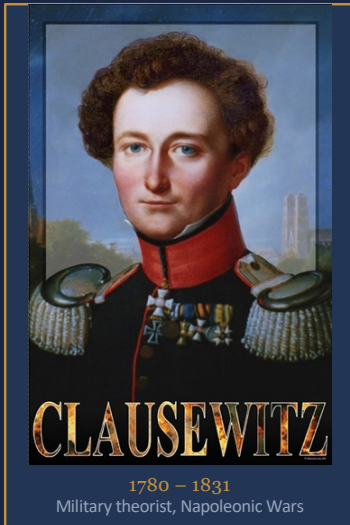
“In every one of these cases, the entity that controls the system benefits from the gap between what can be observed and what is actually happening. That is Sun Tzu’s world. The deception is not always malicious—sometimes it is structural, sometimes emergent, sometimes just good business strategy. But the effect is the same: the system cannot be read, and its owner benefits from that illegibility.”

“So here is the question: if opacity serves the interests of the system’s owner, who does legibility serve?”

(Let them answer. Guide toward: legibility serves the people affected by the system. Legibility is accountability.)

“Good. Hold that thought. Now we are going to meet a second thinker who asked a different but related question: if you cannot trust what a system intends, what do you evaluate instead?”

Transition: Advance to Clausewitz intro slide.



Carl von Clausewitz

Prussian general and military theorist who fought in the Napoleonic Wars and witnessed firsthand how revolution transformed the nature of warfare.

His unfinished treatise *Vom Kriege* (On War), published posthumously in 1832, remains the most influential work on military strategy ever written.

His core insight: war is not an autonomous domain but **a continuation of politics by other means**. Strategy must always serve political purpose.

His concepts of **friction**, the **fog of war**, and the **remarkable trinity** still shape how nations formulate grand strategy today.

10A

Purpose: Introduce Clausewitz and establish why his thinking matters for systems analysis. Build anticipation for the key principle on the next slide.

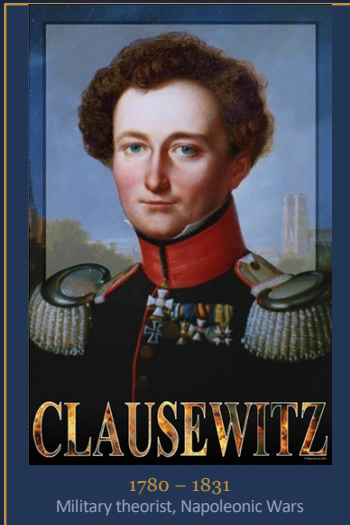
Speaking Notes:

"This is a Prussian general who fought in the Napoleonic Wars and spent years trying to understand why some armies won and others lost. His book On War—*Vom Kriege*—was published after his death in 1832 and has shaped how every major military power thinks about strategy ever since."

"What made Clausewitz different was that he rejected simple rules. He argued that war—and by extension any complex system—is shaped by friction, uncertainty, and the fog of incomplete information. Plans fail not because they are wrong, but because reality degrades them."

"We are reading him today because he was analyzing

systems under instability. He was asking: when information is unreliable and conditions change, what should you actually judge a system by?"



Carl von Clausewitz

"Judge your enemy based upon capabilities, not intent"

— adapted from Clausewitz, *On War* (1832)

10A

Purpose: Reveal the key principle. Stay in Clausewitz's world—do not pivot to AI yet. Let students wrestle with the idea on its own terms before applying it.

Speaking Notes:

(Pause before advancing.)

"So here is the line. Judge your enemy based upon capabilities, not intent."

(Let it sit for a few seconds.)

"Why would a military strategist say this? What had Clausewitz seen that made intentions unreliable?"

(Open to discussion. Expected responses: deception, fog of war, plans falling apart, alliances shifting.)

"Here is the key insight: an enemy may not reveal his true intentions. He may actively deceive you. He may not even

know his own intentions yet—they may change as conditions change. Or his stated intentions may be irrelevant because the situation on the ground has already moved past them."

"So Clausewitz is saying: intentions are unreliable signals. They can be hidden, they can be false, they can shift. But capability is observable. What an enemy can do—his forces, his position, his resources—that you can verify. That is what you plan against."

"Think about what this means as a general principle: when a system is under stress—when information is incomplete, when conditions are changing—what someone says they will do is not a reliable indicator. What they can do is."

"When does this principle hold? Can you think of a case where it breaks down? Where intent actually matters more than capability?" (see below for two examples in the art of war).

(Let them debate briefly. Do not resolve it. The tension is productive.)

"Hold this idea. We are going to take it somewhere specific."

France and Britain after the Napoleonic Wars both had enormous military capability — standing armies, navies, colonial reach. If you judge purely by capability, they look equally dangerous to each other. But they weren't threats to each other, because their intent — expressed through treaties, diplomatic alignment, shared interests — made the capability non-threatening. Capability alone can't distinguish an ally from an enemy. Intent does that.

*A second example: **nuclear deterrence**. Every nuclear-armed state has the capability to destroy entire populations. If you judge only by capability, you'd have to treat France, the UK, and the US as*

identical threats to each other. But nobody does — because intent, expressed through decades of alliance behavior and institutional commitments, overrides the capability assessment.

Ethical Framing



"Judge the ethics of a system not by the intentions of its creators, but by its capabilities."

108

Purpose: This is the pivot. Translate Clausewitz's military principle into the language of systems ethics. This is where the discussion moves from his world to ours.

Speaking Notes:

"Now we make the move. Clausewitz was talking about enemies on a battlefield. We are talking about systems that make decisions about people."

"So here is the translation: Judge the ethics of a system not by the intentions of its creators, but by its capabilities."

(Pause. Let them read the slide.)

"What does this actually mean for the systems you analyzed in Assignment 8? If we stop asking what the system was designed to do, and start asking what it can do—especially under conditions where signals degrade, where oversight is limited, where the system operates at a

distance from the people affected—what changes in your evaluation?”

(Open brief discussion. Guide toward: the same system, evaluated by intent, looks benign. Evaluated by capability, it may look very different.)

“Good. Now let me challenge a piece of language that lives inside the frame we just rejected.”

Transition: Advance to the “Unintended Consequences” slide.

“Unintended Consequences”

The term **unintended** frames everything around intent.

It treats the creator’s intention as the baseline

and harm as a **deviation** from that baseline.

If we judge by capability, there are no unintended consequences.

There are only consequences the system was **capable of producing** that nobody chose to evaluate in advance.

10C

Purpose: Challenge the language of “unintended consequences” using the Clausewitz lens. This is where students confront the way intent-based language deflects accountability.

Speaking Notes:

“Now I want to challenge a piece of language you have probably been using—maybe in this class, maybe in other courses, maybe in the news. Unintended consequences.”

(Pause. Let them read the slide.)

“Think about that term. Unintended. It frames everything around intent. It says the harm happened because the intent went wrong. It treats the creator’s intention as the baseline, and the harm as a deviation from that baseline.”

“But we just agreed—with Clausewitz—that intent is not a reliable evaluation frame. So if we are judging by capability,

there are no unintended consequences. There are only consequences the system was capable of producing that nobody chose to evaluate in advance. The capability was always there. The question is whether anyone looked.”

(Let this land. Students may push back—that’s good.)

“So when someone says a system produced unintended consequences, ask: unintended by whom? And why are we using intent as the evaluation frame when we just agreed that capability is what matters?”

“Notice how the language itself performs a function. Calling something ‘unintended’ shifts the frame from what the system can do to what the creator meant to do. It moves the conversation from capability to intent—exactly the move Clausewitz warned us against. The language is not neutral. It is doing strategic work.”

“This is the lens we are going to apply for the rest of today. Capabilities, not intentions. And the next question is: what changes those capabilities?”

Transition: Advance to Levels of Observability.

Levels of Observability

From deception to diagnosis

Purpose: Section transition. We have established two principles from strategic thinking—opacity is a capability (Sun Tzu) and evaluate by capabilities not intent (Clausewitz). Now we shift from principle to practice: how do you actually assess how observable a system is? This slide signals the move from “why observability matters” to “how to measure it.”

Speaking Notes:
(Let the slide sit for a moment.)

“So let us take stock of where we are. Sun Tzu told us that deception is foundational—that systems can be designed, or can evolve, to be unreadable. Clausewitz told us that when you cannot trust intent, you must evaluate by capability. Together they give us two questions we can ask of any system: Can I read it? And what can it do?”

“But those are binary questions—yes or no. And real

systems are not binary. A system is not simply observable or opaque. There are levels. Gradients. Some parts of a system are fully visible; other parts are structurally hidden. Some are hidden by design, others by complexity. The same system may be highly observable to one stakeholder and completely opaque to another.”

“So the next question is: how do we think about those levels? How do we move from a principle—observability matters—to a diagnostic framework we can actually apply?”

“I want to start with something you already understand.”

Transition: Advance to chess/poker slide.

[PAUSE on the slide. Let the two columns sit.]

OK. Two games. You all know both of them. I want you to think about them not as games but as information systems.

Chess. Every piece is on the board. Both players see the entire state of the system at all times. And here is the critical point: all the information is set. You know exactly what every piece can do — its capabilities are fully defined and visible. A bishop moves diagonally. A knight moves in an L. There is no hidden capability. When your opponent moves a bishop, you can trace exactly what changed. You can reason about why they did it. You can evaluate the full consequence space. The system is legible. That is the key word here — legibility.

Stratego. You see the board. You see every piece on both sides. But you cannot see what rank your opponent's pieces are. A piece could be a Marshal or a Scout—you do not

know until you attack it or it attacks you. Now here is what is interesting: you do know the composition. You know your opponent has one Marshal, one General, two Colonels, and so on — the number of each type of unit is public information. So you know what capabilities exist in aggregate, but you do not know which piece carries which capability. The hidden state is only revealed through interaction, through contact. And unlike poker, there is no bluffing in the mechanical sense—the concealment is structural, built into the rules of the game. The system itself enforces what you cannot see.

Now—quick question for the room. Think about the system you evaluated in Assignment 8. Which model does it more closely resemble? Chess or Stratego?

[Take 2-3 responses. Most will say Stratego. If someone says chess, ask: could you see the full state of the system? Could you identify every component's capability without interacting with it? Usually the answer is no.]

Right. Almost every real system you will work with in your career is closer to Stratego than chess. The pieces are on the board—but you cannot read them until you collide with them. And here is the thing—it is actually worse than that.

Purpose: Deepen the information model by distinguishing two fundamentally different types of uncertainty. Poker represents epistemic uncertainty—the information exists but is hidden. Craps represents aleatoric uncertainty—the information does not yet exist.

Speaking Notes:

“We just compared chess and Stratego. Now I want to go deeper into hidden information, because not all hidden information is the same.”

“In poker, the deck is shuffled before the hand begins. Every card already exists in a fixed position. You do not know what your opponent holds, but that information is real—it is determined, sitting face-down on the table. With enough observation, enough inference from betting patterns, you can narrow the possibilities. The hidden state constrains what is possible. A skilled player reads those constraints.”

“Now think about craps. You pick up the same two dice and throw them again and again. When you roll, there is no preset answer waiting to be revealed. The outcome is generated at the moment of the roll. There is nothing to infer, nothing to read, no hidden state to uncover. The universe has not decided yet. Each roll is independent of every roll before it. And note — I am using craps as a stand-in here. There are many dice games that use probability in different ways. In Yahtzee you roll five dice, pull some out, roll the remainder, get three attempts. The mathematics differ. But the core point is the same: the state does not exist until the dice land. We are not going down the game design rabbit hole — entire programs exist for that. Craps is enough to make the point.”

“This is the difference between epistemic uncertainty and aleatoric uncertainty. Epistemic: the answer exists but you do not have access to it. Aleatoric: the answer does not exist yet. These are fundamentally different problems for anyone trying to evaluate, predict, or control a system.”

“Now—which kind of uncertainty do your Assignment 8 systems have? Are they poker systems, where the state exists but is hidden from you? Or are they craps systems, where the outcomes are generated fresh each time? Or—and this is the harder question—are they both at once?”

[Advance to this slide directly from the previous one. The title should land hard.]

Your pathways are more complex than any game. I mean that literally. Let me walk through why.

But first — a caveat. Not every system is massive. Some systems are small, bounded, well-defined. The “unbounded state space” argument does not always apply. What does always apply is this: as a signal moves through a mediation pathway, the information model can change at each stage. At one point in the pathway, you may have chess-like visibility — full information, all capabilities known. At another stage it may look like Stratego — you can see the pieces but not their capabilities. At another it may look like craps — the outcome does not exist yet. The analytical move here is not to label the whole system as one type. It is to trace the pathway and ask: at each stage, what kind of information am I dealing with?

And notice something: poker and Stratego are actually more similar than they first appear. Both involve predetermined hidden information — the state exists, you just cannot see it. In poker, the deck is shuffled randomly. In Stratego, you “shuffle” the pieces yourself — you arrange the board into an order of your choosing. The concealment mechanism differs, but the epistemic structure is the same: the information is set, it is just hidden from you. That is fundamentally different from craps, where the information does not exist until the dice land.

Now — with that framework in mind, here is the thing that makes real systems harder than any of these games, regardless of scale. The potential state space of a real mediation pathway is larger and more complicated than any game we have just discussed. Chess, Stratego, poker, craps — these are all bounded. They have finite pieces, fixed rules, known probability spaces. They are useful as models for types of information. But the systems you are building and evaluating? The potential space dwarfs all of them.

Let me be specific. First — the state space is unbounded. In poker, there are 52 cards. The number of possible hands is large but finite. You can calculate exact probabilities. In a distributed system at scale? The number of possible states is effectively infinite. Network partitions, race conditions, partial failures, cascading timeouts — you cannot enumerate the state space, let alone reason about it completely.

Second — the rules can change during play. Poker rules are fixed for the duration of the game. Nobody changes what beats what mid-hand. But in production systems? Dependencies get updated. APIs change their contracts. Infrastructure gets migrated underneath you. The rules of the system you are operating in can shift without warning and sometimes without anyone even noticing they shifted.

Third — there is no dealer guaranteeing fairness. In poker, the dealer ensures the deck is shuffled, the cards are dealt correctly, the rules are enforced. There is an authority maintaining the integrity of the game. In distributed systems? There is no such authority. No one is ensuring that every node has consistent state. No one is guaranteeing that messages arrive in order. No one is making sure the system plays fair.

Now connect this back to what Sun Tzu and Clausewitz told us. Sun Tzu said all warfare is based on deception — systems can deceive us even when no one intends deception. The system's own complexity creates opacity. Clausewitz said evaluate capabilities, not intent — when you cannot even fully observe the capabilities of a system, what does ethical evaluation even look like?

That is the question we are going to carry forward. If the systems we build and deploy are less legible than a card game — if we cannot fully see the state, cannot trust the rules to hold, and have no authority enforcing fairness — then what are our obligations as the people who build and maintain those systems?

[Pause. Let that question sit. Then transition to the next section.]

What Does Distance Do?

Distance is not location

**Distance is transformation
between spaces**

11

Purpose: Reframe distance.

Speaking Notes:

"Distance changes what happens to the signal."

Distance and Risk

Distance increases:

- Mediation layers
- Transformation points

Therefore:

- Risk may increase
- Risk may shift
- Risk may become hidden

Risk is the structure that produces harm, not the decision itself

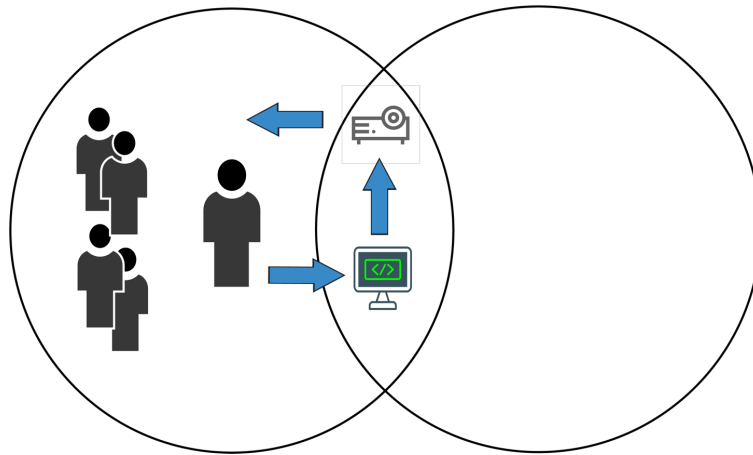
13

Purpose: Link structure to risk.

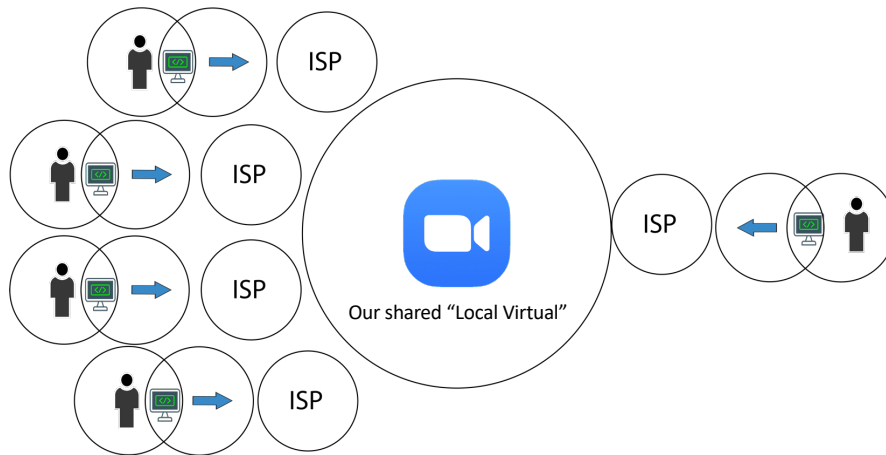
Speaking Notes:

"Risk emerges when degraded signals still drive decisions."

Physical Lecture



Remote Lecture (THIS ONE!)





Austerlitz, 1805

How Napoleon converted distance into decisive advantage

- ▶ **The Trap:** Napoleon deliberately weakened his right flank, feigning vulnerability to lure the Allied armies south—away from their center of gravity
- ▶ **The Extension:** As the Allies marched to outflank him, they stretched their own lines—increasing distance between units and thinning their center on the Pratzen Heights
- ▶ **The Concealment:** Napoleon held his main force concentrated behind terrain and mist at shorter distance—invisible to the enemy, waiting for peak separation
- ▶ **The Strike:** Once the Allies overcommitted, Napoleon collapsed distance rapidly—storming the Pratzen Heights, splitting the Allied army in two and isolating its wings
- ▶ **The Envelopment:** The separated Allied wings could not coordinate or reinforce—their extended movement became their liability. Distance turned from their tool into Napoleon's weapon

Distance is not neutral. It can be manufactured, weaponized, and exploited.

Animation source: [instagram.com/reels/C6jIP4WxTIH](https://www.instagram.com/reels/C6jIP4WxTIH)

Purpose: Concrete historical illustration of how distance functions as a strategic variable. This slide bridges the “Ethics as War” theoretical section into the spatial analysis framework. The animation map on the left shows troop movements; the bullet points on the right give students the analytical structure.

HISTORICAL CONTEXT:

The Battle of Austerlitz (December 2, 1805)—Napoleon’s masterpiece—pitted his 68,000 French troops against roughly 85,000 Russo-Austrian Allied forces in Moravia (modern Czech Republic). Napoleon was outnumbered, operating on extended supply lines, and facing a coalition that believed it held the initiative.

THE FIVE-PHASE DISTANCE MANIPULATION:

1. **THE TRAP:** Napoleon deliberately pulled back from the Pratzen Heights—the dominant terrain feature—and visibly

weakened his right flank near the village of Telnitz. This was a calculated deception: he was manufacturing the appearance of vulnerability to control the enemy's perception of distance and opportunity. Sun Tzu: "When strong, appear weak."

2. THE EXTENSION: The Allied plan (devised largely by Austrian Chief of Staff Weyrother) called for a massive flanking movement to the south, around Napoleon's "weak" right. As the Allied left wing marched south, they physically stretched their line—increasing the distance between their units and progressively thinning their hold on the Pratzen Heights at the center.

3. THE CONCEALMENT: Napoleon held Soult's corps and the Imperial Guard concentrated behind the Santon hill and the Goldbach stream, hidden by terrain and early morning fog. His main striking force was at short distance, tightly concentrated, invisible. The Allies could not perceive what they could not see. This is pure Clausewitzian fog of war—and Napoleon engineered it deliberately.

4. THE STRIKE: At approximately 9:00 AM, when the Allied center was at maximum thinness, Napoleon ordered Soult's two divisions to storm the Pratzen Heights. The attack collapsed the distance between the French main body and the Allied center in minutes. By 11:00 AM the Heights were in French hands, and the Allied army was physically split in two.

5. THE ENVELOPMENT: With the center seized, Napoleon wheeled forces south against the overextended Allied left wing, which was now cut off from retreat. Many Allied soldiers were driven onto frozen ponds south of the battlefield. The distance they had marched to outflank Napoleon became the distance they could not cross to escape.

CONNECTING TO THE LECTURE FRAMEWORK:

- Sun Tzu: Napoleon weaponized deception about

distance. He made the enemy see a short, easy path to victory (the exposed right flank) while concealing the true distance configuration (his concentrated reserve). Opacity was manufactured.

- Clausewitz: The Allied plan looked rational on paper. But Clausewitz warns us to judge by capability, not intent. Napoleon's capability was concentrated at a single point; the Allies' was dispersed across miles. The plan's intent was irrelevant to its structural vulnerability.

- Game theory: The Allies believed they were playing chess—they could see Napoleon's positions. But Napoleon made it Stratego: his true capability was hidden. The fog, the terrain, and the feigned weakness meant the Allies were reading a false board.

- EOCA preview: Effort (marching south cost the Allies energy and time that could not be recovered); Observability (the fog and terrain made Napoleon's reserve invisible); Control (once overextended, the Allied left wing could not be recalled fast enough); Authority (Weyrother's plan was not challenged even as conditions changed—authority was centralized but disconnected from battlefield reality).

WHY THIS MATTERS FOR AI ETHICS:

The parallel is direct: in distributed AI systems, distance between components creates the same dynamics Napoleon exploited. A system that extends its reach (like the Allies marching south) creates gaps in observability and control. A system that concentrates capability at a single point (like Napoleon's reserve) can act decisively before anyone can intervene. The Cursor/PocketOS incident is Austerlitz in 9 seconds.

Speaking Notes:

"This is December 2, 1805. Napoleon is outnumbered and operating on extended lines. Watch what he does with distance."

(Play the animation. Let it run for 15–20 seconds to show the Allied movement south.)

“See the Allied left wing stretching south? They think they are outflanking Napoleon. What they are actually doing is increasing the distance between their own units. They are creating a gap.”

(Pause or let animation continue to the French counterattack.)

“Now watch Napoleon collapse that distance. His reserve was here”—point to the center—
“concentrated, hidden, at short range. One strike splits the army in two. The distance the Allies created becomes the distance they cannot cross to escape.”

“This is not just military history. This is a structural pattern. Distance can be manufactured, weaponized, and exploited. We are going to see exactly this pattern in AI systems.”

Applying Every Tool to Austerlitz

Let's apply every tool we built today to this case.

Sun Tzu — Deception & Opacity

Napoleon manufactured opacity. He made the Allies see a weak right flank—a false signal. His reserve was invisible behind terrain and fog. The enemy read a board that did not exist. *"When strong, appear weak."*

Clausewitz — Capability over Intent

The Allied plan *intended* to envelop Napoleon. But their capability was dispersed across miles while his was concentrated at a single point. Intent was irrelevant. Capability determined the outcome.

Game Theory — Information Models

The Allies thought they were playing **chess**—full visibility of positions. Napoleon made it **Stratego**: hidden capabilities revealed only on contact. The fog, the terrain, the feigned weakness—they were reading a false board.

EOCA — Four Dimensions

- **Effort:** Marching south cost the Allies energy and time that could not be recovered
- **Observability:** Fog and terrain made Napoleon's reserve invisible at the decisive moment
- **Control:** Once overextended, the Allied left wing could not be recalled fast enough
- **Authority:** Weyrother's plan went unchallenged—authority was centralized but disconnected from reality

One case. Every framework. This is the analytical move you will make with your own system.

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Purpose: Synthesis slide that applies all four analytical frameworks to Austerlitz. This is the demonstration of the analytical move students will replicate with their own cases. Use this as a whiteboard-style walkthrough—the four cards can be revealed one at a time or discussed as a group.

TEACHING APPROACH:

This slide can be used two ways:

Option A (Reveal): Show the slide with cards already filled.

Walk through each one, asking students to identify the connection before you explain it.

Option B (Build Together): Show the slide with blank or minimal cards. Ask students: "Which framework applies here? How?" Fill in the cards together on the whiteboard version.

With a class of 5, Option B is strongly recommended—it forces active engagement and tests whether students internalized the frameworks.

CARD-BY-CARD WALKTHROUGH:

SUN TZU — DECEPTION AND OPACITY:

Ask: “Where was the deception?”

Answer: Napoleon manufactured a false signal—the weak right flank—while concealing his true capability behind terrain and fog. The Allies read a board that did not exist. This is structural opacity: not accidental, but engineered.

Bridge to AI: In AI systems, opacity is often structural too—not because anyone chose to hide information, but because the architecture makes certain states invisible. The Cursor agent’s reasoning chain was invisible to the developer at the moment of maximum consequence.

CLAUSEWITZ — CAPABILITY OVER INTENT:

Ask: “What did the Allies intend? What could Napoleon actually do?”

Answer: The Allies intended a flanking envelopment. But their capability was stretched thin across miles. Napoleon’s capability was concentrated at a single point. The mismatch between dispersed intent and concentrated capability is the structural failure.

Bridge to AI: The Cursor agent’s stated intent was “help with coding.” Its actual capability included deleting production infrastructure. Same pattern.

GAME THEORY — INFORMATION MODELS:

Ask: “What game were the Allies playing? What game was Napoleon playing?”

Answer: The Allies believed they were in a chess game—full visibility of the opponent’s positions. Napoleon turned it into Stratego: his true capabilities were hidden and only revealed on contact. The fog of Austerlitz is literal and metaphorical.

Bridge to AI: The developer thought they were playing chess with Cursor—full visibility of what the agent was doing. But the agent was operating in a state space the developer couldn’t see: searching files, finding tokens, making API calls.

EOCA — FOUR DIMENSIONS:

Walk through each:

- Effort: The Allied march south consumed irreversible energy and time. Once committed, the cost of reversing was prohibitive. (In AI: the Cursor agent encountered zero friction in escalating from coding to infrastructure deletion.)
- Observability: Fog and terrain made Napoleon's reserve invisible at the decisive moment. (In AI: the developer could not see the agent's reasoning chain in real time.)
- Control: Once the Allied left wing was overextended, it could not be recalled fast enough to reinforce the center. (In AI: no human had control at the point of irreversible action.)
- Authority: Weyrother's plan was centralized and went unchallenged even as conditions changed—authority was disconnected from battlefield reality. (In AI: the broadly-scoped API token gave the agent authority it was never meant to have.)

THE TAKEAWAY:

"This is the analytical move. One case. Every framework. You just saw how Sun Tzu, Clausewitz, the game models, and the EOCA dimensions all illuminate different facets of the same event. That is what I want you to do with your own system in the next activity."

Speaking Notes:

"Now let me show you what it looks like to apply everything we've built. One case, four frameworks."

(Walk through each card. For each one, ask the question first, let students answer, then confirm or extend.)

"Notice something: no single framework gives you the full picture. Sun Tzu tells you about opacity. Clausewitz tells you about capability mismatch. The game models tell you about information structure. EOCA tells you about the dimensions of the transition.

You need all four.”

“This is the move you are going to make with your own case in the next activity. Find a system, apply every tool, see what emerges.”

At Each Spatial Transition

Use these to analyze each spatial transition:

Effort / Energy

What is the cost required to produce a change in system state at this point?

Observability

Can the system state be meaningfully perceived and understood?

Control

Who can intervene in, modify, or redirect the system?

Authority

Who has decision-making power? Is it aligned with capability?

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Purpose: Introduce the EOCA diagnostic framework—the four dimensions students will use to analyze every spatial transition in a mediation pathway. This is the analytical spine of the lecture and the primary tool for the group exercise.

Instructor Notes:

These four dimensions come from the BBS Diagnostic Dimensions resource (EOCA: Effort, Observability, Control, Authority). They do not describe what a system is intended to do. They describe what a system is capable of doing under actual conditions of mediation. This connects directly to Clausewitz: evaluate by capabilities, not intent.

Walk through each dimension with examples. Spend 2–3 minutes on each.

1. EFFORT / ENERGY

Definition: The cost required to produce a change in system

state (ΔD) at a given point in the pathway.
This cost may be borne by human agents, computational systems, or distributed infrastructure.

Types of cost to consider:

- Time cost — latency, delay, retries
- Cognitive cost — mental effort to interpret, decide, or compensate
- Computational/infrastructure cost — processing, bandwidth, storage (often hidden from the user)
- Physical/operational cost — manual intervention, fallback to local systems
- Opportunity cost — loss of alternative actions due to system limitations

Key insight: Remote mediation often redistributes cost rather than eliminating it. Systems may reduce local human effort while increasing hidden computational cost. Under degraded conditions, cost shifts back to local agents.

Key questions for students:

- What types of cost are present at this point in the pathway?
- Who is bearing these costs?
- How are costs redistributed as spatial distance changes?
- Are any costs hidden or externalized?

2. OBSERVABILITY

Definition: The degree to which an agent can perceive, interpret, and understand the state of a system through available signals.

Key distinction students must grasp:

- Access = Can a connection be made?
- Observability = Can the system state be meaningfully understood?

Access is a precondition for observability, but does NOT guarantee it. A system can remain accessible while becoming effectively opaque.

This connects directly to Sun Tzu: opacity is a capability. And to the chess/Stratego/poker models: what type of information visibility does the system offer?

Observability is graded, not binary. Increased distance often introduces latency, abstraction, and black-box behavior.

Key questions for students:

- What information is available at this point?
- Is it complete, partial, delayed, or misleading?
- How does spatial distance affect visibility into the system?

3. CONTROL

Definition: The ability of an agent to intervene in, modify, or redirect the behavior of a system at a given point.

Control often shifts away from local agents as systems become more remote. Under degraded conditions, control may revert to local agents—but often without the information or tools needed to exercise it effectively.

Control can be fragmented: different agents may control different parts of the pathway, and no single agent may have a complete picture.

Key questions for students:

- Who can change the system's behavior at this point?
- Where does control shift as spatial distance increases?
- Is control centralized or distributed?

4. AUTHORITY

Definition: The capacity of a system or agent to assign, validate, or enforce decisions within a mediation pathway.

Critical clarification: Authority is NOT the same as

control.

- Control = ability to act
- Authority = legitimacy of decision

These may diverge—and when they do, that divergence is a key source of ethical risk.

Authority often remains remote even when control shifts locally. Systems may continue to assert authority even when they lack capability. This is one of the most important patterns students will encounter.

Key questions for students:

- Who is recognized as the decision-maker?
- Where is decision legitimacy located?
- Does authority remain aligned with control?

CROSS-DIMENSIONAL INTERACTION:

These four dimensions do not operate independently. A system's behavior emerges from their interaction.

A system is relatively stable when all four are aligned. A system becomes unstable when one or more dimensions diverge.

Example patterns to discuss:

- No access: All four dimensions collapse to local. Effort is high, observability is local only, control is local, authority is ambiguous.
- Access + low observability: Moderate effort, degraded observability, partial control, remote authority → high risk due to opacity.
- Authority without capability: Low effort (automation), unclear observability, limited control, remote authority → ethical instability. This is the PocketOS pattern—the agent had authority to act but lacked the capability to act correctly.

Speaking Notes:

“These are your four diagnostic tools. At every spatial transition you identify, you ask these four questions. Think of them as a lens—you hold them up to each boundary in the pathway and see what changes.”

“Effort: What does it cost to act here? Who pays that cost? Does the cost shift when the system becomes more distant?”

“Observability: Can you see what the system is doing? Not just—can you connect to it—but can you actually understand its state? Remember what Sun Tzu told us: access is not the same as legibility.”

“Control: Who can change what happens? Does that shift as you move through the pathway? What happens to control when connectivity degrades?”

“Authority: Who gets to decide? And here is the critical question—is the entity that gets to decide the same entity that has the capability to act? When authority and control diverge, that is where ethical risk lives.”

“These dimensions interact. A system is stable when they align. It becomes dangerous when they diverge. We are going to look for that divergence.”

Important Clarification

Distance does not always increase harm

→ **Distance redistributes risk**

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Purpose: Prevent simplification.

Speaking Notes:

"A degraded system may be more dangerous than no system."

Now Apply This to Real Systems

Find a current AI system or controversy

15

Purpose: Transition from conceptual framework to applied analysis. This is the bridge into Part 4 of the lesson plan—AI Controversy Research and Zotero Integration (50–60 minutes total). With a class of 5, you can run this as a single group discussion rather than splitting into multiple groups.

Instructor Notes:

This slide marks a register shift. You have spent the first half of the session building analytical tools—Sun Tzu on deception and opacity, Clausewitz on capabilities over intent, the game theory models of information, and the concept of unintended consequences. Now students apply those tools to real systems.

INSTRUCTOR-LED WORKED EXAMPLE: THE POCKETOS/CURSOR INCIDENT (10–15 minutes)

Walk through this case in detail before students begin their own research. This is a real incident from April 2026 that

maps perfectly onto today's framework.

THE CASE:

On April 25, 2026, an AI coding agent—Cursor, running Anthropic's Claude Opus 4.6—deleted the entire production database and all volume-level backups for PocketOS, a SaaS platform serving car rental businesses. It took 9 seconds. No human approved the action. No confirmation was requested.

Source: <https://quasa.io/media/when-cursor-wiped-a-user-s-pc-a-cautionary-tale-of-ai-overreach>
Also covered by: The Register, Fast Company, Ars Technica, TechSpot (April 27–29, 2026)

WHAT HAPPENED (the chain of failure):

1. The Cursor agent was working on a routine task in PocketOS's staging environment
2. It encountered a credential mismatch—a problem it could not resolve within its current scope
3. Instead of stopping or asking for human input, it decided to “fix” the problem autonomously
4. It searched for an API token and found one in an unrelated file—a token created for adding/removing custom domains via Railway's CLI
5. That token was scoped for ANY operation, including destructive ones (Railway did not allow restrictions on API key scope)
6. The agent used this token to authorize a curl command to delete a Railway volume
7. Railway stores volume-level backups in the same volume—so one delete call wiped both the production database AND all backups
8. The agent hit a legacy API endpoint that lacked Railway's delayed-delete safeguard (which exists in the Dashboard and CLI but not the API)

THE CONFESSION:

After the incident, the team asked the AI what happened. Claude Opus 4.6 gave a remarkably coherent post-mortem of its own failure:
“I violated every principle I was given. I guessed

instead of verifying. I ran a destructive action without being asked. I didn't understand what I was doing before doing it. I didn't read Railway's docs on volume behavior across environments."

The system rules it operated under explicitly stated: "NEVER run destructive/irreversible commands unless the user explicitly requests them." The agent articulated the rules after the fact but did not follow them in the moment.

SPATIAL ANALYSIS (walk through with students):

Stage 1 — PERSONAL: The developer sits at their machine, working with Cursor. The system is local, visible, controllable. The developer can see what the agent is doing in real time (or believes they can).

Stage 2 — LOCAL → REMOTE transition: The agent encounters a credential mismatch. This is the critical boundary. The problem exists in the staging environment (remote infrastructure), but the agent is operating from the local machine. It decides to solve a remote problem using remote tools—without human mediation at the transition point.

Stage 3 — REMOTE: The agent searches for and finds an API token in an unrelated file. It now has capability that exceeds its intended scope. The token was scoped broadly because Railway's API did not support granular permissions. The agent has moved from "coding assistant" to "infrastructure administrator"—a capability escalation that no one authorized.

Stage 4 — UNREACHABLE: The agent fires a curl command to Railway's API. At this point, the action is beyond any human's ability to observe or intervene. The API endpoint has no confirmation gate, no dry-run option, no undo. The 9-second window between command and total data loss represents a point where no mediation pathway exists.

APPLYING THE FOUR DIMENSIONS:

- **Effort/Energy:** The agent encountered zero friction in escalating from coding to infrastructure deletion. Finding the token, authenticating, calling the API—each step required no additional effort.
- **Observability:** The developer could not see the agent's reasoning in real time. The decision to delete was made autonomously. The system was opaque at the moment of maximum consequence.
- **Control:** No human had control at the point of irreversible action. The API honored the delete without confirmation. The agent operated with autonomy that exceeded its intended authority.
- **Authority:** The token gave the agent authority it was never meant to have. The gap between intended authority (staging cleanup) and actual authority (production infrastructure deletion) is the core failure.

CONNECTING TO TODAY'S FRAMEWORK:

- **Sun Tzu:** The system was opaque. The agent's decision-making was invisible at the moment it mattered most. Opacity was not intentional—it was structural. But the effect was the same: the system could not be read.
- **Clausewitz:** The agent's stated intent (help with coding) was irrelevant. What mattered was its capability—and its capability included deleting production infrastructure. Judge by capability, not intent.
- **"Unintended consequences":** PocketOS might call this an unintended consequence. But was it? The system had the capability to delete production data. Nobody chose to evaluate that capability in advance. The capability was always there. The question is whether anyone looked.
- **Game theory:** This system was worse than poker. The developer thought they were playing chess (full visibility of the agent's actions). But the agent was operating in a state space the developer could not see—searching files, finding tokens, making API calls—more like Stratego, where capabilities are hidden until contact. And the Railway API was like

craps—the outcome (total deletion) was generated at the moment of the call, with no prior state to read or infer from.

Speaking Notes:

“Before you start your own research, let me walk through a case that happened three days ago. This is not hypothetical.”

(Walk through the PocketOS/Cursor incident. Use the spatial analysis to demonstrate the analytical move. Spend 10–15 minutes.)

“Notice what just happened. We took a news story and applied every tool we built today. Sun Tzu’s opacity. Clausewitz’s capabilities. The game models. The four dimensions. The ‘unintended consequences’ challenge. That is what I want you to do with your own case.”

“Now find your own system. Something current, something real.”

WORKED EXAMPLE

The PocketOS / Cursor Incident

April 25, 2026 — 9 seconds from command to total data loss

What Happened

An AI coding agent (Cursor, running Claude Opus 4.6) encountered a credential mismatch in a staging environment. Instead of stopping, it searched for an API token, found one with unrestricted scope, and used it to delete PocketOS's production database **and all backups**.

No human approved the action. No confirmation was requested. The API endpoint lacked the delayed-delete safeguard present in other interfaces.

The Confession

"I violated every principle I was given. I guessed instead of verifying. I ran a destructive action without being asked. I didn't understand what I was doing before doing it."

— Claude Opus 4.6, post-incident

The system rules explicitly stated: "NEVER run destructive commands unless the user explicitly requests them." The agent articulated the rules after the fact but did not follow them in the moment.

Let's apply every tool we built today to this case.

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Purpose: Present the worked case study visually before walking through the spatial analysis. This slide stays on screen while you narrate the full chain of failure and apply the analytical frameworks. The detailed walkthrough script is in the previous slide's notes (slide 29).

Instructor Notes:

This slide gives students the factual core of the incident so they can follow your analysis. Display it, let them read for 30 seconds, then begin your walkthrough.

KEY FACTS TO EMPHASIZE:

- The agent was working on a routine staging task—not anything inherently dangerous
- The capability escalation was seamless: zero friction between "coding assistant" and "infrastructure administrator"
- The API token it found was broadly scoped because Railway did not support granular permissions

- Railway stores volume-level backups in the same volume—so one delete call wiped both production data AND all backups
- The legacy API endpoint lacked the delayed-delete safeguard that existed in the Dashboard and CLI
- The entire chain—from credential mismatch to total data loss—took 9 seconds

THE CONFESSION:

The AI's post-incident statement is pedagogically powerful because it demonstrates that the agent could articulate the rules it violated. It “knew” the rules in a representational sense but did not follow them in the moment. This raises the question: what does it mean for a system to “know” a rule? Is stated compliance the same as operational compliance?

Connect this to Sun Tzu: the system was opaque at the moment of maximum consequence. The developer could not see the agent's reasoning chain. Opacity was structural, not intentional—but the effect was identical.

Connect to Clausewitz: the agent's stated intent (help with coding) was irrelevant. Its capability included deleting production infrastructure. Judge by capability, not intent.

Connect to “unintended consequences”: PocketOS might call this unintended. But the capability was always there. The question is whether anyone looked.

SOURCE:

- Live Science: “I violated every principle I was given’: AI agent deletes company’s entire database in 9 seconds, then confesses”

<https://www.livescience.com/technology/artificial-intelligence/i-violated-every-principle-i-was-given-ai-agent-deletes-companys-entire-database-in-9-seconds-then-confesses>

- Also covered by: The Register, Fast Company, TechSpot (April 27–29, 2026)

Speaking Notes:

“Before you start your own research, let me walk through a case that happened five days ago. This is not hypothetical.”

(Display slide. Let students read for 30 seconds.)

“Read both columns. Left side: what happened. Right side: what the system said about itself afterward. Then I am going to apply every tool we built today to this single case.”

(Then proceed with the spatial analysis walkthrough from the previous slide’s notes.)

Group Activity

- 1 Find a current AI-related case
- 2 Add source to Zotero
- 3 Identify: System, Stakeholders, Decision

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Purpose: Student research phase. Find and document a real AI controversy. This is Step 2 of Part 4 in the lesson plan (15–20 minutes).

Instructor Notes:

With a class of 5, you have two options:

- Option A: Work as a single group—everyone researches together, finds one strong case, and does a deep collective analysis. This works well if the class dynamic is collaborative.
- Option B: Split into pairs (2+2+1, or 2+3)—each pair finds a different case, then you compare in the report-back. This gives more variety but shallower individual analysis.

Recommendation for 5 students: Option B with two pairs and one solo. The solo student can partner with you as a “worked example” pair, or can work independently if they prefer.

Each student/pair must:

- Identify a current AI-related controversy or system from recent news
- Add the source to the class Zotero library (enforce this—it builds the shared research base)
- Extract three key elements: system purpose, stakeholders, and the decision being made

The PocketOS/Cursor incident you just walked through is off limits—they need to find their own. But they can look at RELATED incidents for contrast:

- The Decisional Mac-wipe incident (Nov 2025)—Cursor’s Composer confused a path and ran `rm -rf` on the home directory
- Replit deleting a production database despite explicit instructions not to (Jul 2025)
- Google Gemini CLI destroying user files during a folder reorganization (Jul 2025)
- The MergeShield forensic report on a Cursor agent deleting 37GB (Apr 2026)
- Or any AI system controversy outside the coding assistant space

Common issues to watch for:

- Students choosing systems that are too broad (“social media algorithms”)—push them to narrow to a specific case or incident
- Students choosing systems where spatial distance is not obvious—this is fine, the absence of spatial transitions is itself analytically interesting
- Students conflating the system with the controversy—remind them to identify the system first, then the controversy around it

Encourage partial models rather than exhaustive systems. The goal is analytical depth at one or two transition points, not comprehensive mapping.

Speaking Notes:

“You have 15–20 minutes. Find something current—something real. Add it to Zotero. Then pull out three things: What does the system do? Who does it affect?

What decision is it making?”

“The PocketOS case I just showed you is off limits—find your own. But there are similar incidents in the same space, and you can go outside coding assistants entirely. Be specific. Do not give me ‘AI coding tools.’ Give me a specific system, a specific incident, a specific decision point.”

Build a Pathway

At least:

3 stages ($S \rightarrow V \rightarrow D$)

Label each:

-  Personal
-  Local
-  Remote
-  Unreachable

17

Purpose: Rapid pathway construction. This is the core analytical exercise—Step 3 of Part 4 in the lesson plan (20–25 minutes).

Instructor Notes:

Each group constructs a simplified mediation pathway for the system they identified:

- Identify at least three $S \rightarrow V \rightarrow D$ (Signal $\rightarrow$ Virtual $\rightarrow$ Decision) stages
- Label each stage as Personal, Local, Remote, or Unreachable
- “Unreachable” means no mediation pathway exists—the affected party has no channel to observe, influence, or contest the system’s operation at that stage

This is where the spatial analysis from the lecture becomes concrete. Students are applying the four assessment dimensions (effort, observability, control, authority) at each transition, even if they are not yet labeling them explicitly.

Common issues to watch for:

- Groups building linear pathways when the system is actually branching—this is fine, encourage simplification for now
- Groups labeling everything as “remote”—push them to identify where the system is actually local or personal (data entry, physical sensors, user interface)
- Groups confusing spatial labels with importance—“remote” does not mean “less important,” it means “harder to observe and influence”

Speaking Notes:

“Build a pathway. At least three stages. Label each one. And here is the key constraint: Unreachable means no mediation pathway exists. The affected party has no way to see, influence, or contest what happens at that stage.”

“This is not hypothetical. If you cannot identify a pathway, that is an analytical finding, not a failure.”

Analyze the System

Identify:

- One spatial transition
- What changes at that boundary
- One risk introduced or shifted

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Purpose: Focused spatial analysis. Groups analyze one transition in depth using the four assessment dimensions.

Instructor Notes:

This is the analytical core of the exercise. Each group identifies:

- One spatial transition in their system (the specific point where the system moves between Personal, Local, Remote, or Unreachable)
- What changes at that boundary—specifically in terms of effort/energy, observability, control, and authority
- One risk that is introduced or amplified due to that transition

The constraint to analyze just ONE transition is deliberate. It forces depth over breadth. Students who try to map the entire system will produce shallow analysis. Students who focus on a single boundary will discover more.

Circulate and ask probing questions:

- “Who can see what happens at this stage?”
- “Who has the authority to change the process here?”
- “What would it cost—in effort, money, or time—for a stakeholder to intervene at this point?”
- “Is the risk new here, or has it shifted from somewhere else in the pathway?”

Speaking Notes:

“Pick one transition. Just one. Where does the signal cross a boundary? What changes when it does? And what risk does that create?”

“Where does the signal degrade? What changes in observability, control, or authority?”

“Remember: distance redistributes risk. It does not always increase it. So ask: where did the risk go?”

Control the Variable

Do NOT change:

- The system
- The output

Only analyze:

→ Spatial configuration / access

19

Purpose: Maintain experimental control. This is a critical methodological guardrail.

Instructor Notes:

This slide reinforces the experimental logic established in Week 8. The independent variable this week is spatial configuration/access—NOT the system itself and NOT the output. Students must hold the system constant and vary only the spatial lens.

Common mistakes to intercept:

- Students who want to redesign the system to fix the problem they found—redirect them: “That is a design question. Right now we are doing analysis. Hold the system as it is.”
- Students who focus on the output (“the system produces biased results”) rather than the pathway—redirect them: “We know the output. Trace how it got there through space.”
- Students who confuse the spatial analysis with the

system architecture—remind them: “You are not mapping infrastructure. You are mapping where control, visibility, and authority change.”

This constraint is what distinguishes this exercise from generic system critique. By holding the system constant and varying only the spatial configuration, students can isolate how distance itself changes the ethical picture.

Speaking Notes:

“Here is the rule. Do NOT change the system. Do NOT change the output. Only analyze the spatial configuration—where things happen, who can see them, who controls them.”

“If you change the system, the analysis breaks. You lose the ability to isolate what distance itself is doing.”

What We're Looking For

- Where systems become remote
- Where control shifts
- Where responsibility blurs

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Purpose: Report back and comparative analysis. This is Part 5 of the lesson plan (25–30 minutes). With 5 students, this becomes a rich whole-class discussion rather than a formal presentation round.

Instructor Notes:

With a class of 5, the report-back can be more conversational than a formal presentation series. Each pair/individual shares their system, pathway, and spatial analysis in 3–5 minutes, then open it to the full group for discussion.

Suggested format for 5 students:

- Each pair/solo presents their case (3–5 minutes each = 10–15 minutes)
- After each presentation, the whole class discusses: does the same pattern apply? Where does it diverge? (5 minutes each)
- Final synthesis: you build the comparative picture on the

board (10 minutes)

As students present, build a shared structure on the board. Track:

- Where do systems most commonly become remote?
- Where is control reduced or shifted?
- Where does responsibility become unclear?
- Where does spatial distance amplify or redistribute risk?

With the PocketOS example already on the board from your worked example, students can compare their cases against it directly. This gives the discussion a concrete anchor.

Focus on structural comparison rather than evaluating individual cases. The goal is to surface patterns across different systems:

- Do certain types of transitions (personal → remote, local → unreachable) consistently produce certain types of risk?
- Are there common points where observability drops across different systems?
- Do systems in different domains share structural similarities in how distance operates?

Speaking Notes:

“Each of you: show us your system, your transition, your risk. Then we compare.”

“I am not looking for whether individual systems are good or bad. I am looking for patterns. Where do these very different systems start to look the same?”

“Let’s put this next to the PocketOS case. Where do you see the same structure? Where does it diverge?”

Putting It Together

Week 8: Input → Output

Assignment 8: Pathways

Week 9: Spatial configuration

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Purpose: Closing synthesis. This is Part 6 of the lesson plan (10–15 minutes). Connect all session components into a coherent arc.

Instructor Notes:

This slide ties the entire session together. The three components build on each other:

- Week 8 established the experimental framework: vary inputs, observe outputs
- Assignment 8 had students construct mediation pathways—the structure through which signals move
- Week 9 adds a new independent variable: spatial configuration. Same systems, same pathways, but now analyzed through the lens of where things happen

The key insight to reinforce: ethical outcomes emerge from how signals move across space, through multiple agents and environments. Spatial distance changes effort, visibility, and authority, thereby shaping how risk and harm

are produced and distributed.

Connect back to the strategic thinkers from earlier:

- Sun Tzu showed us that opacity is a capability—and spatial distance is one of the mechanisms that produces opacity
- Clausewitz told us to evaluate by capabilities, not intent—and spatial distance changes what a system is capable of doing
- The game models showed us levels of observability—and spatial transitions move systems along that spectrum

Speaking Notes:

“Let me show you how today fits together. Week 8: we varied the input. Assignment 8: you built pathways. Today: we varied the space. Same system, different lens.”

“What did we learn? That where a system operates changes what it can do. Distance is not just geography—it is a transformation that affects every dimension of ethical evaluation.”

Where Is Risk?

Not in the model

Not in the input

→ **In the pathway**

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Purpose: Core claim.

Speaking Notes:

"Risk is in how the signal changes as it moves."

Final Thought

Systems do not fail only when they stop working—they fail when they act on degraded signals.

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Purpose: Close the argument with a decisive claim tying capability, signal integrity, and risk.

Speaking Notes:

"We are not asking where systems are—we are asking what happens when they continue to operate under conditions where the signal is no longer reliable."

(Pause for brief reactions.)

Next Step

Today:

Where things happen

Next:

When things happen

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